The effects of fruit juices and fruits on the absorption of iron from a rice meal.

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The effects of the chemical composition of fruit juices and fruit on the absorption of iron from a rice (Oryza sativa) meal were measured in 234 parous Indian women, using the erythrocyte utilization of radioactive Fe method. The corrected geometric mean Fe absorptions with different juices varied between 0.040 and 0.129, with the variation correlating closely with the ascorbic acid contents of the juices (rs 0.838, P less than 0.01). Ascorbic acid was not the only organic acid responsible for the promoting effects of citrus fruit juices on Fe absorption. Fe absorption from laboratory 'orange juice' (100 ml water, 33 mg ascorbic acid and 750 mg citric acid) was significantly better than that from 100 ml water and 33 mg ascorbic acid alone (0.097 and 0.059 respectively), while Fe absorption from 100 ml orange juice (28 mg ascorbic acid) was better than that from 100 ml water containing the same amount of ascorbic acid (0.139 and 0.098 respectively). Finally, Fe absorption from laboratory 'lemon juice' (100 ml orange juice and 4 g citric acid) was significantly better than that from 100 ml orange juice (0.226 and 0.166 respectively). The corrected geometric mean Fe absorption from the rice meal was 0.025. Several fruits had little or no effect on Fe absorption from the meal (0.013-0.024). These included grape (Vitis vinifera), peach (Prunus persica), apple (Malus sylvestris) and avocado pear (Persea americana). Fruit with a mild to moderate enhancing effect on Fe absorption (0.031-0.088) included strawberry (Fragaria sp.) (uncorrected values), plum (Prunus domestica), rhubarb (Rheum rhaponticum), banana (Musa cavendishii), mango (Mangifera indica), pear (Pyrus communis), cantaloup (Cucumis melo) and pineapple (Ananas comosus) (uncorrected values). Guava (Psidium guajava) and pawpaw (Carica papaya) markedly increased Fe absorption (0.126-0.293). There was a close correlation between Fe absorption and the ascorbic acid content of the fruits tested (rs 0.738, P less than 0.0001). There was also a weaker but significant correlation with the citric acid content (rs 0.55, P less than 0.03). Although this may have reflected a direct effect of citric acid on Fe absorption, it should be noted that fruits containing citric acid also contained ascorbic acid (rs 0.70, P less than 0.002).(ABSTRACT TRUNCATED AT 400 WORDS)